TR-M203
Ram Representations and Certificates for the Sharp Isotropic and Entropy Problem

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Abstract

The target is the sharp simplex upper bound for the affine-invariant isotropic constant, together with the stronger entropy inequality for every log-concave Lebesgue density. M203 asserts both statements in all dimensions. Its geometric cone reduction is elementary once the entropy inequality is available; its substantive analytic burden lies in affine normalization of a Gaussian transport, noncommuting matrix estimates, scalar enclosures, and limiting rigidity. This report reads that chain at source level and separates it from what is independently proved here.

The local Ram corpus directly represents the same moment determinant, its physical Gram metric, and its finite polynomial certificates. We prove a mass-coupled relative moment bound strictly stronger than the uncoupled determinant/volume estimate, an exact signed cone-deficit identity, a finite Gaussian covariance Gram certificate with a positive tail, an entropy certificate including the missing energy integral, stable residual propagation for the scalar ODEs, and a quantitative finite-jet curl/commutator bound. A sharper chaos remainder is proved as an exact spectral identity and yields a conditional strengthening of M203's final estimate, with exact coefficients 1689103/407000000 and 29/13200 and three retained nonnegative terms. The reconstructed source certificate passes 201 exact checks; an additional checker for this report passes 133. These results do not establish the global entropy or simplex conjectures. They specify useful, domain-correct representations and smaller independently checkable tasks.

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1 Disposition and exact scope

M203 remains a level-5 applicability case in the user's calibration: the corpus treats the *same core functional*, not merely an analogous transform or a peripheral change of coordinates. This does not mean the corpus already proves the manuscript's theorem. In particular, a certificate for a specified convex body, a specified parameter box, or a specified transported density is not a theorem for every body or every log-concave density.

The main independently established contributions of this report are as follows.

1. The cone lift has exactly the signed deficit $\Delta_{n+1}(f_K) = \frac{1}{2}\log(S_n/F(K))$, with all dimensional factors and the Mellin measure explicit (Section 5).
2. A mass-coupled Loewner certificate saves a factor u/l over a standard uncoupled relative enclosure. An exact rational square example separates the two tests (Section 6).
3. The Gaussian resolvent covariance is a physical row Gram matrix. Truncating its Hermite representation leaves a positive semidefinite tail with an explicit norm bound (Section 7).
4. Scalar approximation errors can be propagated in their stable directions, with rational damping factors and explicit joining errors (Section 10).
5. A full-density entropy certificate requires both an augmented moment matrix and the potential-energy integral. We give the exact formula and an interval test (Section 11).
6. Small zeroth and first curl jets control matrix commutators under explicit derivative assumptions (Section 12).
7. An exact chaos identity retains a positive remainder discarded in the source. Its entropy consequence is explicitly conditional on the upstream matrix and scalar estimates (Section 13).

A result established here has a proof below. This is not a priority claim: the moment, Mellin, Schur-complement, Bernstein, and Hermite ingredients are classical. The sharper coupling and the other refinements are deductions made explicit for this assessment.

1.1 Two exact targets

For a full-dimensional compact convex body $K \subset \mathbb{R}^n$, let

$$V = |K|, \qquad \mu = \frac{1}{V}\int_K x\,\mathrm{d}x, \qquad \Sigma = \frac{1}{V}\int_K (x-\mu)(x-\mu)^T\,\mathrm{d}x.$$

Set

$$F(K) = \frac{\det\Sigma}{V^2} = L_K^{2n}, \qquad S_n = \frac{(n!)^2}{(n+1)^{n+1}(n+2)^n}. \tag{1.1}$$

The sharp body target is $F(K) \le S_n$, with equality exactly for simplices. No restriction to polytopes, rational vertices, finitely many facets, or bounded computational complexity is part of this target.

The functional target is, for every log-concave probability density f with respect to Lebesgue measure on $\mathbb{R}^m$,

$$\Delta_m(f) := h(f) - m - \frac{1}{2}\log\det\mathrm{Cov}(f) \ge 0, \qquad h(f) = -\int_{\mathbb{R}^m} f\log f. \tag{1.2}$$

The asserted equality class is the invertible affine images of products of independent mean-one one-sided exponential laws. A full-dimensional log-concave density has positive covariance and the integrability needed here; Section 14 explains the source's coercivity route. Lower-dimensional measures require a separate intrinsic formulation and are not covered by the displayed determinant.

Both quantities are affine invariant. For $Y = AX + c$, A invertible, $h(Y) = h(X) + \log|\det A|$ and $\det \mathrm{Cov}(Y) = (\det A)^2 \det \mathrm{Cov}(X)$, so Δ_m is unchanged. In the body problem volume transforms by $|\det A|$ and the same cancellation gives $F(AK + c) = F(K)$. Nothing in this invariance implies the sign of Δ_m.

1.2 Logical status

There are three different assertions throughout this report:

Source assertion.
M203 claims the all-dimensional inequalities and complete equality classification. A source label identifies where that claim or ingredient occurs; it is not an independent certification of it.

Independent result.
A theorem proved in this report uses only its displayed hypotheses and standard finite-dimensional, integral, or Hilbert-space arguments.

Conditional consequence.
An improvement follows if specifically named M203 inequalities are valid on their stated domains. Such a consequence does not validate its premises.

The distinction is especially important for the finite scalar certificate: exact polynomial arithmetic verifies a polynomial statement, but transferring it to Gaussian hazard functions also requires anchors, residuals, correct error transport, positive denominators, and complete tail coverage.

2 Source chain, provenance, and reading depth

The catalogue row was read at `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/00-Applicability-Catalogue/applicability.json`, entry M203 (family 101), referring to source commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. M203 is a public OpenAI manuscript; the local files below are the downloaded copy of that public source, not part of the confidential Ram corpus. The pinned public PDF is `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-sharp-entropy-bound-and-the-simplex-inequality-for-isotropic-constants-October-5-2026/isotropic-simplex.pdf`. The present reading is deeper than the original screening row. All ten files listed below were read in full, including mathematical proof prose, rather than only abstracts or keyword matches. This authoring-stage reading is distinct from the independent source audit and its reconstructed 201-check finite certificate, subsequently read and rerun locally. The replay covers the finite assertions enumerated in Section 16; it does not formally verify every analytic step or the cited external transport theorems.

Define the exact source prefix M by

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/A-sharp-entropy-bound-and-the-simplex-inequality-for-isotropic-constants-October-5-2026/build/manuscript/
```

Every filename in the following table is relative to this exact prefix. Labels are the source's LaTeX labels, not invented theorem numbers.

File	Exact source anchors	Reading and proof boundary
main.tex	Document assembly, abstract, input order	Full read. Confirms the asserted global scope; not evidence of correctness.
introduction.tex	intro:isotropic-constant, intro:simplex, intro:simplex-bound, intro:entropy, intro:entropy-bound	Full read. Targets, constants, equality statements and claimed consequences were compared with the body and entropy formulations here.
transport.tex	tr:normalization, tr:energy, tr:Jacobian-equation, tr:surplus	Full read. Traced degree normalization, elliptic energy and third-tensor surplus. Regularity imports and all analytic estimates are not independently certified.
matrix.tex	mat:covariance, mat:deficit-reduction, mat:commutator, mat:scalar-absorption, mat:scalar-pair, mat:strict-remainder	Full read. Traced matrix ordering and final chaos reduction. Covariance and final spectral slack are independently rederived below. The independent audit also reviewed the ordered proof and supplied a passing exact algebra stress test. This is not a formal verification of the complete analytic chain.
scalar.tex	sc:bounds, sc:substitution-table	Full read. Traced scalar bounds, absorption minorants, and endpoint/pair costs. The independent replay checks all 36 substitution corner comparisons and selected further rational bounds. The real-line secant argument remains analytically reviewed, not machine-formalized.
approximation.tex	geo:coercivity, geo:approximation	Full read. Moreau regularization, common tails and convergence route checked structurally; global conclusion still depends on the smooth inequality.
rigidity.tex	lem:linear-jacobian, lem:spectral-lipschitz	Full read. Local curl implication rederived and quantified below. The global limiting equality argument is described with its dependencies.
geometry.tex	eq:body-cone-entropy, eq:body-cone-covariance, eq:body-cone-determinant, eq:body-cone-bound	Full read. Cone identities and determinant constants independently rederived here; equality transfer is conditional on functional rigidity.
certificate.tex	cert:recurrence, cert:residuals, cert:global-errors, cert:aux-errors, cert:Bernstein, cert:group-bounds	Full read. The independently reconstructed checker, rerun locally, passes all 44 group margins, 20 joins and the further finite checks listed in Section 16. This does not formalize every analytic tail argument.
references.bib	Complete bibliography	Full read for provenance. A bibliography entry does not establish that its theorem's hypotheses have been checked in this application.

The source certificate mentions `verification/verify_scalar.py`. That file was not present in the downloaded M203 source inventory examined for this report. It was not executed here. The two independently reconstructed checkers and this report's additional refinement checker are distinct from that missing source verifier.

2.1 Direct corpus sources

The principal local source, read in full, is

> `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-344-astra.tex`.

Its exact anchors are `sec:target`, `thm:moment`, `eq:transport`, `thm:radius`, `thm:cert`, `cor:relative`, `prop:dual`, `thm:inclusion`, `thm:continuity`, `lem:simplex`, `thm:bernstein`, `lem:variation`, and `thm:reflection`. It explicitly distinguishes certificates for actual bodies from arbitrary positive moment matrices and does not claim a global simplex proof.

The following are direct, domain-specific foundations, not merely words shared by the two corpora.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`: the sections on what a Ram representation means and the concrete polynomial construction, `eq:moments` and `thm:transfer`. The relevant polynomial construction and transfer passages were read. They identify a finite, unisolvent polynomial representation and the exact invertible coordinate change on that space. They do not supply positivity outside the represented space.
2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`: `v16c05:powerbernstein`, `v16c05:conversionproof`, `v16c05:hullbound`. The conversion, convex-hull proof and subdivision discussion were read directly. They apply to actual polynomial coefficient enclosures on a specified interval or box.
3. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`: `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:residual-identity`, `v21c01:gram-equivalence`. The defining statements and proofs were read. A finite synthesis bank uses its physical Gram metric; unresolved residuals are not removed by renaming coordinates.
4. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume2/chapters/ch01-master-atoms.tex`: `v2:c01:mellin-def`, `v2:c01:mellin-log`, `v2:c01:mellin-inverse`, `v2:c01:mellin-energy`. The Mellin/log-coordinate and weighted Fourier statements were read. Inversion requires the stated integrability and line of convergence. The cone computation below only needs absolutely convergent integrals.
5. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`: `v21c02:hankel`, `v21c02:localizer`, `v21c02:atomic-certificate`. Relevant finite moment/positivity passages were read. A positive moment sequence or atomic representing measure is not automatically the uniform measure of a convex body.

The local discovery maps `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-foundations-map.txt`, `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-core-map.txt`, and `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/corpus-applications-map.txt` were read and used for navigation. A map entry is not substituted for reading the theorem cited in this report. Nonlinear integral-equation contraction and analytic-continuation theorems are not invoked: no required contractive map, invertibility bound,

or continuation domain has been established for the critical transport normalization or matrix estimate.

3 Public primary literature and what it does not settle

Only public mathematical terms and public bibliographic metadata were used for the literature checks. No private corpus text was submitted to a search engine or external service. The following is a bounded current literature check as of October 7, 2026, not a claim of exhaustive discovery of every contemporaneous preprint.

Klartag and Lehec's published 2025 paper proves a dimension-independent upper bound on isotropic constants [3]. Its Theorem 1.2 and introductory discussion distinguish that result from the sharp fixed-dimensional simplex conjecture. Thus solving the ordinary slicing problem does not supply S_n or its equality class.

Fradelizi and Marín Sola formulate the functional entropy/isotropic conjecture and examine its relationship with the body conjecture [4]. In the public version inspected, Conjecture 3(iii), Proposition 5.2 and Lemma 5.2 are the relevant anchors. In particular the functional statement in dimension $n+1$ implies the body statement in dimension n, while the converse route uses higher-dimensional body statements. This is not a same-dimension equivalence and is not evidence that either sharp global statement has been proved.

Melbourne, Nayar and Roberto prove the sharp one-dimensional minimum-entropy statement with fixed variance, with one-sided exponential equality [5]. Their Theorem 1.1 was checked. It supplies a genuine scalar benchmark for (1.2), not an automatic tensorization for arbitrary dependent log-concave laws in higher dimensions.

Klartag's work links sharp isotropic-constant assertions to Mahler volume questions [6]. The public abstract's stated implication was checked; the entire proof of that implication was not independently audited here. This report does not infer a Mahler theorem from M203. Kipp's 2026 paper on shadow systems gives necessary conditions on local maximizers using decomposability [7]; its published introductory statement was read. Such a necessary condition is not a classification of all maximizers. TR-item-344's facet calculations belong to this narrower variational setting, not a replacement for the global entropy argument.

The classical transport and functional-calculus imports in M203 include Brenier transport, Caffarelli-type contraction/regularity, divided differences for matrix functions, Gaussian chaos identities and Moreau approximation. This report checks how these are being used and proves the specific elementary consequences stated below. It does not claim a fresh independent audit of every imported theorem. The source bibliography is the record of those imports; the references at the end of this report separate the primary public works consulted from confidential local source documents.

4 Moment and physical-Gram rederivation

Let $z(x) = (1, x^T)^T$ and use unnormalized moments

$$b = \int_K x \, \mathrm{d}x, \qquad Q = \int_K xx^T \, \mathrm{d}x, \qquad H(K) = \int_K z(x)z(x)^T \, \mathrm{d}x = \begin{pmatrix} V & b^T \\ b & Q \end{pmatrix}. \tag{4.1}$$

Full dimensionality implies $H \succ 0$: the integral of the square of a nonzero affine polynomial cannot vanish on the interior of K.

Proposition 4.1 (Exact augmented-moment identity)**.** *For every body in* (1.1)*,*

$$\det H = V^{n+1} \det \Sigma, \qquad F(K) = \frac{\det H}{V^{n+3}}. \tag{4.2}$$

For $y = Ax + c$*,* $d = |\det A|$ *and* $T = \left(\begin{smallmatrix} 1 & 0 \\ c & A \end{smallmatrix}\right)$*,*

$$H(AK + c) = d\, T H(K) T^T, \qquad \det H(AK + c) = d^{n+3} \det H(K). \tag{4.3}$$

Proof. The Schur complement of V in H is $Q - bb^T/V = V\Sigma$. Hence $\det H = V \det(V\Sigma) = V^{n+1} \det \Sigma$. The change of variables gives $z(Ax+c) = Tz(x)$ and $\mathrm{d}y = d\,\mathrm{d}x$, proving the congruence. Since $(\det T)^2 = d^2$, its determinant scales by $d^{n+1}d^2$. □

This is TR-item-344, `thm:moment` and `eq:transport`, rederived with the exact exponent. The exponent is $n + 3$, not $n + 1$.

For affine polynomials $p_u(x) = u^T z(x)$,

$$\|p_u\|^2_{L^2(K)} = u^T H u.$$

The physical metric is H, not the Euclidean length of the coordinate vector u. Replacing z by a nonsingular finite polynomial bank Ez changes the Gram matrix to EHE^T, changes coefficients contragrediently, and preserves the represented function and its norm. This is the applicable content of the polynomial Ram transfer and physical Gram sources in Section 2.1. It adds a usable representation; it does not create a new positivity theorem.

4.1 Absolute and relative certificates

Suppose an actual body's moments satisfy

$$|V - V_0| \le a, \quad \|b - b_0\| \le b_*, \quad \|Q - Q_0\|_{\mathrm{op}} \le c.$$

For a unit vector (s, v),

$$|(s, v)^T (H - H_0)(s, v)| \le as^2 + 2b_*|s|\|v\| + c\|v\|^2.$$

Consequently

$$\|H - H_0\|_{\mathrm{op}} \le \rho_* = \frac{a + c + \sqrt{(a - c)^2 + 4b_*^2}}{2}. \tag{4.4}$$

This is the sharp algebraic block estimate in TR-item-344, `thm:radius`. Sharpness as a matrix estimate does not assert geometric realizability of the extremizing blocks.

If $V_- \le V \le V_+$, $V_- > 0$, then

$$F(K) \le \frac{\det(H_0 + \rho_* I)}{V_-^{n+3}}.$$

A lower determinant bound using $H_0 - \rho_* I$ requires that matrix to be positive semidefinite. A positive determinant alone is not enough: an indefinite matrix can have positive determinant. Relative Loewner bounds are particularly natural because their constants survive the congruence (4.3). Section 6 improves the usual relative certificate by using the fact that the mass is also an entry of H.

4.2 Realization and geometry cannot be omitted

For example the measure giving mass $1/2$ to each of $-1, 1$ has $H = \mathrm{diag}(1, 1)$. A uniform interval of length one has centered second moment $1/12$, not one. Positive H with a positive constant entry therefore does not certify a uniform convex body.

There is a useful geometric substitute. If $K_- \subset K \subset K_+$, then $H(K_-) \preceq H(K) \preceq H(K_+)$, because every quadratic form of the difference is the integral of an affine square over a nonnegative measure. Covariance itself does not have this simple inclusion monotonicity: normalization and recentering interfere. TR-item-344's `thm:inclusion` works with the augmented moments for exactly this reason. Rational simplex integration can supply the moments of verified inner and outer polytopes; it cannot certify an unverified triangulation or a missing geometric inclusion.

5 The cone, Mellin factorization, and exact signed gap

Let

$$C_K = \{(tx, t) : x \in K,\ t \geq 0\}, \qquad f_K(y, t) = \frac{e^{-t}}{n!V} \mathbf{1}_{C_K}(y, t).$$

The cone is convex and f_K is log-concave. The map $(x, t) \mapsto (tx, t)$ has Jacobian t^n for $t > 0$.

Proposition 5.1 (Mellin moments of a cone)**.** *For a multi-index $\alpha \in \mathbb{N}^n$, an integer $j \geq 0$ and $\Re\lambda > 0$,*

$$\int_{C_K} y^\alpha t^j e^{-\lambda t} \, \mathrm{d}y \, \mathrm{d}t = \frac{\Gamma(n + |\alpha| + j + 1)}{\lambda^{\,n+|\alpha|+j+1}} \int_K x^\alpha \, \mathrm{d}x. \tag{5.1}$$

All these integrals are absolutely convergent.

Proof. Compactness of K bounds $|x^\alpha|$. After the change of variables, the integral factors into its K moment and $\int_0^\infty t^{n+|\alpha|+j} e^{-\lambda t} \, \mathrm{d}t$. The gamma integral gives the identity; the same estimate with $\Re\lambda$ proves absolute integrability. □

For the Mellin convention in `v2:c01:mellin-def`,

$$\mathcal{M}g(s) = \int_0^\infty g(t) t^{s-1} \, \mathrm{d}t = \int_{\mathbb{R}} g(e^u) e^{su} \, \mathrm{d}u, \qquad t = e^u.$$

In (5.1), $s = n + |\alpha| + j + 1$. The bare cone measure is $t^n \, \mathrm{d}t = e^{(n+1)u} \, \mathrm{d}u$, not $\mathrm{d}u$. This weight is necessary for every dimensional constant. On a line $s = c + i\omega$, the associated weighted L^2 measure is $t^{2c-1} \, \mathrm{d}t$; a Mellin Plancherel assertion must respect that measure. No analytic continuation is required to compute the moments above, and analytic continuation would not establish a sign inequality.

Theorem 5.2 (Exact signed cone deficit)**.** *For every full-dimensional compact convex $K \subset \mathbb{R}^n$,*

$$\boxed{\Delta_{n+1}(f_K) = \frac{1}{2} \log \frac{S_n}{F(K)} = n \log \frac{L_{\mathrm{simplex}}}{L_K}}. \tag{5.2}$$

The identity is unconditional; its nonnegativity is not asserted here.

Proof. Let X be uniform on K, and independently let T have the gamma law with shape $n+1$ and rate one. The preceding Jacobian proves (TX, T) has density f_K. We have

$$\mathbb{E}T = n+1, \quad \mathbb{E}T^2 = (n+1)(n+2), \quad \operatorname{Var} T = n+1.$$

Write $\mu = \mathbb{E}X$. The determinant-one shear $(Y, T) \mapsto (Y - \mu T, T)$ changes neither entropy nor covariance determinant. After the shear, the cross covariance is zero and the covariance is

$$\operatorname{diag}\big((n+1)(n+2)\Sigma,\ n+1\big). \tag{5.3}$$

Also $-\log f_K(TX, T) = T + \log(n!V)$ almost surely, so $h(f_K) = n+1+\log(n!V)$. Subtract $n+1$ and half the logarithm of the determinant in (5.3). The result is

$$\log(n!V) - \tfrac{1}{2}\log\big((n+1)^{n+1}(n+2)^n \det\Sigma\big),$$

which is (5.2) by (1.1). □

This rederives the source's `eq:body-cone-entropy` through `eq:body-cone-bound`. It makes the strongest direct Ram connection particularly transparent: the body determinant certificate and the entropy certificate of its cone concern *exactly the same signed quantity*, not approximately similar objectives.

If the entropy inequality and its equality classification in dimension $n+1$ are established, this identity proves the body inequality in dimension n. Equality forces the cone support to be an affine image of an orthant. Its unique extreme point is the origin, so the affine translation must vanish. All generating rays then have positive last coordinate; their section at height one is a simplex. Conversely, for simplex vertices $v_0, \dots, v_n$, the matrix with columns $(v_i, 1)$ has absolute determinant $n!V$ and sends the positive orthant to C_K. Its last coordinate is the sum of the orthant coordinates, giving the exponential product law exactly. The necessity direction here is conditional on the functional equality classification, not a separate proof of it.

6 A sharper mass-coupled relative certificate

The relative estimate in TR-item-344, `cor:relative`, combines a determinant interval and an independent volume interval. The volume is actually $e_0^T H e_0$, which allows a strictly sharper conclusion.

Theorem 6.1 (Mass-coupled Loewner bound)**.** *Let $H, H_0 \succ 0$ be $(n+1)$-by-$(n+1)$ matrices, and put*

$$V = e_0^T H e_0, \quad V_0 = e_0^T H_0 e_0, \quad \mathcal{F}(H) = \det H / V^{n+3}.$$

Suppose $0 < l \le u$, $lH_0 \preceq H \preceq uH_0$, and

$$r_- \le r := V/V_0 \le r_+, \qquad 0 < r_- \le r_+.$$

The volume interval may be intersected with $[l, u]$. Then

$$\frac{l^n}{r_+^{n+2}} \le \frac{\mathcal{F}(H)}{\mathcal{F}(H_0)} \le \frac{u^n}{r_-^{n+2}}. \tag{6.1}$$

For fixed admissible r, these algebraic bounds are attained by suitable positive definite matrices. This attainment need not be geometrically realizable as uniform convex-body moments.

Proof. Whiten H_0: set $B = H_0^{-1/2} H H_0^{-1/2}$. Then $lI \preceq B \preceq uI$. The vector $v = H_0^{1/2} e_0/\sqrt{V_0}$ is a unit vector, with $v^T B v = r$. Choose an orthonormal basis beginning with v, and write

$$B = \begin{pmatrix} r & z^T \\ z & C \end{pmatrix}, \qquad S = C - zz^T/r.$$

For $w \in \mathbb{R}^n$,

$$w^T S w = \min_{a \in \mathbb{R}} (a, w)^T B (a, w) \geq l\|w\|^2.$$

On the other hand $S \preceq C \preceq uI$. Thus $l^n \leq \det S \leq u^n$, while $\det B = r \det S$. Dividing by r^{n+3} proves the bound for fixed r, and monotonicity in r gives (6.1). In this orthonormal basis $B = \operatorname{diag}(r, l, \ldots, l)$ and $B = \operatorname{diag}(r, u, \ldots, u)$ attain the lower and upper bounds. □

With only $lH_0 \preceq H \preceq uH_0$, the new bound is

$$\frac{l^n}{u^{n+2}} \leq \frac{\mathcal{F}(H)}{\mathcal{F}(H_0)} \leq \frac{u^n}{l^{n+2}}.$$

The uncoupled estimate would give l^{n+1}/u^{n+3} and u^{n+1}/l^{n+3}, respectively. Each endpoint improves by the factor u/l. All quantities used in this comparison are invariant under the simultaneous affine congruence (4.3); this is a physical moment improvement, not a choice of favorable coordinates.

6.1 An exact rational separation

Take the unit-area centered square in dimension two,

$$H_0 = \operatorname{diag}(1, 1/12, 1/12), \quad \mathcal{F}(H_0) = 1/144, \quad S_2 = 1/108.$$

Suppose an actual body has the relative enclosure $(21/22)H_0 \preceq H \preceq (23/22)H_0$. The new upper multiplier is

$$\frac{(23/22)^2}{(21/22)^4} = \frac{256036}{194481} < \frac{4}{3}, \qquad 3(256036) = 768108 < 777924 = 4(194481).$$

It certifies $F(K) < 1/108$. The uncoupled upper multiplier is

$$\frac{(23/22)^3}{(21/22)^5} = \frac{5888828}{4084101} > \frac{4}{3},$$

so that test alone is inconclusive. There is no assertion that every matrix in this neighborhood is realizable. The statement is: every actual body whose moments satisfy this neighborhood is certified by the new test. The exact square is one such body. The comparison is a genuine improvement in the sufficient criterion, not a new extremal classification.

6.2 Polynomial parameter families

Suppose K_θ is a verified family of rationally parameterized polytopes on a box Θ, with a valid triangulation, nonzero oriented simplex volumes of fixed sign, and no missing or overlapping interiors. When the vertex coordinates are polynomial and these signs are fixed, simplex integration gives polynomial $V(\theta)$, $b(\theta)$ and $Q(\theta)$. The target reduces to

$$P(\theta) = S_n V(\theta)^{n+3} - \det H(\theta) \geq 0. \tag{6.2}$$

For rational rather than polynomial coordinates, denominators must first be certified nonzero with fixed sign before clearing them. A determinant formula cannot repair a failure of the geometric parameterization.

For one variable, if $p(t) = \sum_{k=0}^{d} a_k t^k$, then

$$p(t) = \sum_{i=0}^{d} \beta_i \binom{d}{i} t^i (1-t)^{d-i}, \quad \beta_i = \sum_{k=0}^{i} a_k \frac{\binom{i}{k}}{\binom{d}{k}}. \tag{6.3}$$

Indeed

$$t^k = \sum_{i=k}^{d} \frac{\binom{i}{k}}{\binom{d}{k}} \binom{d}{i} t^i (1-t)^{d-i},$$

by the binomial theorem after extracting t^k. The tensor product gives the multivariable identity. These basis functions are nonnegative and sum to one, so a common nonnegative lower bound on their coefficients proves the polynomial nonnegative throughout the box. This is precisely the classical coefficient equivalence and hull bound of volume 16 and TR-item-344, `thm:bernstein`.

A negative coefficient is inconclusive: for instance $(t - \frac{1}{2})^2$ has degree-two Bernstein coefficients $(1/4, -1/4, 1/4)$. Exact subdivision or degree elevation may help, but a zero-margin case does not enjoy a universal strict-positivity termination argument. Even a complete finite certificate for (6.2) proves only the family covered by its geometric hypotheses.

7 Gaussian covariance as a physical Gram certificate

Let γ_m be standard Gaussian measure, let $N = -\Delta + x \cdot \nabla$ be the nonnegative Ornstein–Uhlenbeck operator, and let $R = (I + N)^{-1}$. On the degree-d Hermite chaos, $N = d$ and $R = (1 + d)^{-1}$. For a vector map $F \in H^1(\gamma_m; \mathbb{R}^m)$, set $J = DF$. Symmetry of J is not needed in the next result.

Theorem 7.1 (Exact covariance Gram and positive truncation tail)**.** *Write $J_{ir} = \sum_\alpha c_{ir,\alpha} h_\alpha$ in the orthonormal Hermite basis, and put $C = \mathrm{Cov}_{\gamma_m}(F)$. Then*

$$C_{ij} = \sum_r \langle J_{ir}, R J_{jr} \rangle_{L^2(\gamma_m)} = \sum_{r,\alpha} \frac{c_{ir,\alpha} c_{jr,\alpha}}{1 + |\alpha|}. \tag{7.1}$$

Let C_N retain only terms with $|\alpha| \le N$, where $N \in \{0, 1, 2, \ldots\}$ is a nonnegative integer. Then

$$0 \preceq C - C_N \preceq \frac{\|J - \Pi_{\le N} J\|^2_{L^2(\gamma_m; F)}}{N + 2} I. \tag{7.2}$$

If $J \in H^1$, a further computable sufficient bound is

$$0 \preceq C - C_N \preceq \frac{\|\partial J\|^2_{L^2(\gamma_m; F)}}{(N+1)(N+2)} I. \tag{7.3}$$

Here F in a norm subscript denotes the Frobenius matrix norm.

Proof. For degree-d components $F_{i,d}$ and $F_{j,d}$, $d \ge 1$, Gaussian integration by parts gives $\sum_r \langle \partial_r F_{i,d}, \partial_r F_{j,d} \rangle = d \langle F_{i,d}, F_{j,d} \rangle$. Differentiation lowers the degree to $d - 1$, on which R acts by $1/d$. Summation over d proves the first identity; orthogonality gives its coefficient form. Density and the H^1 norm extend the calculation from polynomials to the stated domain.

For each (r,α) let $c_{r,\alpha}$ be the column vector with entries $c_{ir,\alpha}$. The tail is the sum of positive matrices $c_{r,\alpha}c_{r,\alpha}^T/(1+|\alpha|)$. Its operator norm is at most its trace, which is bounded by the right side of (7.2). Finally,

$$\sum_{r,i,|\alpha|>N} |c_{ir,\alpha}|^2 \le \frac{1}{N+1}\sum_{r,i,\alpha} |\alpha|\,|c_{ir,\alpha}|^2 = \frac{\|\partial J\|^2}{N+1}.$$

□

The integer cutoff is substantive. For $F(x)=x^2-1$ in one dimension, $J=2x$ and $C=2$. If the same denominators were used literally with $N=1/2$, then $C_N=0$ but the proposed upper bounds would be $8/5$ and $16/15$, both smaller than 2. For a real-valued cutoff one must replace N by $\lfloor N\rfloor$ in the denominators. The theorem above uses integer truncation order throughout.

Thus C_N is the Gram matrix of the finite coefficient columns $c_{r,\alpha}/\sqrt{1+|\alpha|}$. This is a direct physical Gram realization of M203's `mat:covariance`, with an explicit unresolved-space contribution. If $C_N \succ 0$, then

$$\log\det C_N \le \log\det C \le \log\det(C_N+\varepsilon_N I)$$

for either valid ε_N above. If the coefficient integrals are only enclosed, their numerical Gram error must also be charged; a sampled coefficient array cannot be treated as the exact projection. If C_N is singular, this particular lower logarithmic certificate does not apply. None of these covariance statements computes entropy without an additional integral or the transport Jacobian.

For a nonorthonormal finite Ram bank with synthesis map $\mathcal{J}$, the coordinate metric is $\mathcal{J}^*\mathcal{J}$. Replacing orthonormal Hermite coordinates by such a bank requires retaining that metric and the omitted-space residual, as in `v21c01:residual-identity`. A coordinate Euclidean residual alone does not justify (7.2). Likewise a compressed operator is not the full N unless its domain and projection errors are accounted for.

8 The actual transport and matrix chain

This section isolates the difficult source mechanism. It is not a replacement of the full argument by a Gram determinant. Unless otherwise specified, expectations are with respect to γ_m, and $\tau(M)=\mathbb{E}\,\mathrm{tr}\,M$.

8.1 Smooth domain and nonlinear affine normalization

The initial source domain is

$$f=e^{-V},\quad \int e^{-V}=1,\quad V \text{ smooth},\qquad 0<c_-I\preceq D^2V\preceq c_+I.$$

The Brenier map $F=\nabla\phi$ pushes γ_m to f, and its Jacobian $J=D^2\phi$ is positive definite with upper and lower bounds obtained from the cited transport regularity. Define scalar functions

$$t(a)=-\log\Phi(-a),\quad q(a)=t'(a)=\frac{\varphi(a)}{\Phi(-a)},\quad d(a)=q(a)-a,\quad k(a)=\frac{t(a)}{q(a)},\quad b=k',\quad l=b'. \tag{8.1}$$

Here φ,Φ are the standard Gaussian density and distribution function. Direct differentiation gives

$$q'=qd,\qquad b=1-kd,\qquad l=k-q+ab,\qquad q-k=ab-l. \tag{8.2}$$

The source establishes $q : \mathbb{R} \to (0, \infty)$ increasing and onto, $0 < q' < 1$, $q'' \geq 0$, and $d > 0$. It defines $A = q^{-1}(J)$ by spectral calculus.

The essential normalization is $\mathbb{E}A = 0$, not $\mathrm{Cov}(f) = I$. In `tr:normalization` the source varies $V_\theta = (1-\theta)|y|^2/2 + \theta V(y)$, then considers the Brenier Jacobian to the P-affine image of this law for $P \succ 0$. The zero equation is

$$\mathcal{F}(\theta, P) = \mathbb{E}q^{-1}(J_{\theta,P}) = 0.$$

The argument uses continuity, compactness of the entire zero set, and nonzero Brouwer degree at the Gaussian endpoint, where the map is $q^{-1}(P)$. It is not a Banach fixed-point construction.

Some of the compactness mechanism can be stated precisely. At a zero, convexity of q applied to each vector spectral measure yields $\mathbb{E}J \succeq q(0)I$. The covariance identity and its constant-chaos term give $PC_\theta P \succeq (\mathbb{E}J)^2 \succeq q(0)^2 I$, which controls P^{-1}. Lower spectral bounds for J, the Lipschitz bound on q, $\mathbb{E}A = 0$, and the upper transport bound control $\mathbb{E}\,\mathrm{tr}\, J$ and then P. Establishing continuity of the Jacobian map and these bounds uniformly along the homotopy is part of the source proof burden. A numerical root for one P, or ordinary covariance whitening, would not discharge it.

8.2 Energy and an indispensable Hessian constraint

The linearized operator in `tr:energy` is

$$\mathcal{L}u = \mathrm{tr}(J^{-1}D^2u) - \nabla V(F) \cdot \nabla u = e^{|x|^2/2}\,\mathrm{div}\Big(e^{-|x|^2/2}J^{-1}\nabla u\Big). \tag{8.3}$$

The equation for the Jacobian is

$$\mathcal{L}J_{ij} = -\delta_{ij} + (JD^2V(F)J)_{ij} + \mathrm{tr}(J^{-1}J_iJ^{-1}J_j), \qquad J_i = \partial_i J. \tag{8.4}$$

The source uses cutoffs and a trace energy bound to obtain $\partial J \in L^2$, hence $A, k(A) \in H^1$ on the bounded spectral range of the smooth stage.

The third derivative tensor of ϕ is fully symmetric. After diagonalizing A pointwise, derivatives of matrix functions are given by divided differences:

$$(\partial_r v(A))_{ij} = v[a_i, a_j](\partial_r A)_{ij}.$$

The source rotates all tensor indices, including the derivative index; it does not differentiate eigenvectors. This distinction is essential at repeated eigenvalues, where divided differences extend continuously but a chosen eigenbasis need not be differentiable.

In `tr:surplus` the Hessian compatibility and the log-slope estimates yield

$$\mathbb{E}\,\mathrm{tr}\,k(A)^2 \geq \|\partial A\|^2 - \|\partial k(A)\|^2 + \mathcal{C}, \tag{8.5}$$

where, writing $Z_r = \partial_r A$,

$$\mathcal{C} = \mathbb{E}\sum_{i,j,r} r_{ij}\delta_{ij}^2 (Z_r)_{ij}^2, \qquad \delta_{ij} = |\log q(a_i) - \log q(a_j)|,$$

$$r_{ij} = \frac{1 + m_0(\min(a_i, a_j))}{16}\left(1 + \frac{\delta_{ij}^2}{30} + \frac{\delta_{ij}^4}{1680}\right).$$

The function m_0 is the source's six-interval lower bound, with values

$$0, \quad 19/100, \quad 38/100, \quad 68/100, \quad 68/100, \quad 68/100$$

on intervals separated by $-3, -2, -1, 1/2, 4$. This is a quantitative analytic inequality. An arbitrary symmetric matrix field or an arbitrary finite Gram array does not satisfy the Hessian constraints from which it is derived.

8.3 Covariance completion and ordering

Let $C = \mathrm{Cov}(f)$, and put

$$\mathcal{D} = 2m + \log\det C - 2h(f) = -2\Delta_m(f).$$

The transport change of variables and $\log q(a) = -a^2/2 - \frac{1}{2}\log(2\pi) + t(a)$ give the exact identity

$$\mathcal{D} = \tau(A^2 - 2t(A)) + \log\det C + m. \tag{8.6}$$

For $U = q(A) - (I+N)k(A)$, interpreted in H^{-1}, let W be its row Gram matrix in that Hilbert space. Expanding the Gram quadratic form, using $qk = t$, gives

$$C = W + \mathbb{E}\left(2t(A) - k(A)^2 - \sum_r K_r^2\right), \qquad K_r = \partial_r k(A). \tag{8.7}$$

The H^1–H^{-1} pairing is the correct domain. No assumption $Nk(A) \in L^2$ is necessary. If $U = u - \partial^* v$, the dual Cauchy–Schwarz estimate gives

$$\mathrm{tr}(SW) \le \|u\|_S^2 + \|v\|_S^2 \quad (S \succ 0 \text{ constant}),$$

where $\|M\|_S^2 = \mathbb{E}\,\mathrm{tr}(M^T S M)$, with a sum over r for families.

Decompose the normalized field as

$$X_r = \mathbb{E}(x_r A), \qquad A = \sum_r x_r X_r + Y, \qquad B = \sum_r X_r^2.$$

Then Y has only chaoses of degree at least two. Write $G = \partial Y$, $g = \partial N^{-1} Y$, and $V_0 = \|G\|^2 - \|Y\|^2$. The source fixes

$$\alpha = \frac{3}{40}, \quad s = \frac{40}{37}, \quad S = (s^{-1}I + \alpha B)^{-1}, \quad T = S - I,$$
$$H_* = -T(B - I) = \alpha S(B-I)^2 \succeq 0.$$

The symbol H_* is used here to avoid confusing this matrix with the body moment matrix H. The log-determinant supporting inequality is

$$\log\det C + m \le \mathrm{tr}(SC) - \log\det S.$$

Together with (8.5) and (8.7), it gives the source's `mat:deficit-reduction`. Its remaining terms contain T, functions of A, Y, and derivatives. In general these matrices do not commute.

Specifically, in an eigenbasis of A, the defects

$$D_r = K_r - X_r b(A), \qquad L_r = \partial_r b(A) - X_r l(A)$$

need not be symmetric. Both identities $K_r = X_r b + D_r = b X_r + D_r^T$ are used. Replacing $D_r^T D_r$ by $D_r D_r^T$ inside a trace with a noncommuting weight is not a valid unrecorded simplification. The weighted completion also has S on the left and $p(A)^{-1/2}$ on the right, with

$$p = b + kl - \frac{l^2}{\alpha} > 0.$$

Its positivity comes from a scalar estimate, not from positivity of S.

8.4 An independently rederived noncommuting estimate

The following finite matrix lemma supplies a concrete part of the analytic chain that does not depend on any Gaussian scalar table.

Proposition 8.1 (Compensated commutator bound)**.** *Let X_r be real symmetric matrices, $B = \sum_r X_r^2$, $\alpha > 0$, $s > 0$, $S = (s^{-1}I + \alpha B)^{-1}$, and $T = S - I$. For every real symmetric b, arbitrary real matrices w_r, and*

$$P_T = \operatorname{tr}(TbBb - TBb^2),$$

one has

$$-P_T - 2\operatorname{tr}\left([T,b]\sum_r X_r w_r\right) - \operatorname{tr}\left(T\sum_r w_r^T w_r\right) \le \left(1 + \frac{s}{8}\right)\sum_r \|w_r\|_F^2. \tag{8.8}$$

The same estimate can be integrated whenever all terms are integrable.

Proof. Diagonalize S. Because $B = \alpha^{-1}(S^{-1} - s^{-1}I)$, entrywise expansion gives

$$P_T = \frac{1}{2\alpha}\|S^{-1/2}[T,b]S^{-1/2}\|_F^2.$$

Set $F_0 = \sum_r X_r w_r$. Completing the square in the skew-symmetric matrix $S^{-1/2}[T,b]S^{-1/2}$ bounds the first two terms in (8.8) by $2\alpha\|\operatorname{skew}(S^{1/2}F_0S^{1/2})\|_F^2$.

Write $S_i = su_i$, $0 < u_i \le 1$. The rows of $X = (X_1, \dots, X_m)$ are orthogonal because $XX^T = B$ is diagonal. Resolve each column of the stacked w_r along these normalized rows and their orthogonal complement. If z_{ij} is the component of column j along row i, then $(F_0)_{ij} = \sqrt{B_i}\, z_{ij}$; zero rows contribute nothing. Combining the skew square with the last term of (8.8), the pair (z_{ij}, z_{ji}), $i < j$, has matrix

$$\begin{pmatrix} 1 - su_iu_j & -s\sqrt{u_iu_j(1-u_i)(1-u_j)} \\ -s\sqrt{u_iu_j(1-u_i)(1-u_j)} & 1 - su_iu_j \end{pmatrix}.$$

For $z = \sqrt{u_iu_j}$, $(1-u_i)(1-u_j) \le (1-z)^2$. Its largest eigenvalue is therefore at most $1 + s(z - 2z^2) \le 1 + s/8$. Diagonal and orthogonal-complement components have coefficient at most one. Summing proves the claim. □

This independently checks the finite-dimensional mechanism in `mat:commutator`. In particular it does not assume A, $b(A)$, B, and S commute. The proof establishes this lemma, not the subsequent scalar absorption or the transport surplus.

8.5 Where the scalar inputs enter

The source introduces

$$\beta = \frac{7}{25}, \quad c = \frac{3}{2}, \quad \gamma = \frac{3}{4}, \quad \epsilon = \frac{3}{20}, \quad \kappa = \frac{49}{100}, \quad \rho = \epsilon I + \kappa S/s,$$

and auxiliary functions $f_* = k(1+b)$, $h_0 = f_* - ck - 2(b-\beta)l/\alpha$. The absorption statement `mat:scalar-absorption` bounds the logarithmic cost and the remaining H_*-weighted scalar terms by

$$\frac{69}{500}\|Y\|^2 - \frac{1}{1000}\Omega_B, \qquad \Omega_B = \sum_i \omega(\lambda_i(B)),$$

where

$$\omega(\lambda) = \frac{3(\lambda-1)^2}{37+3\lambda}\left(\frac{3}{20} + \frac{49}{100}\frac{37}{37+3\lambda}\right)\min\left\{\frac{13}{100}, \frac{32}{25\sqrt{\lambda}}\right\}. \tag{8.9}$$

At $\lambda = 0$ the minimum uses $13/100$. It vanishes exactly at one and grows like $(24/125)\sqrt{\lambda}$ at infinity. The source proves the absorption through scalar quadratic minorants of $p(a)$ averaged against vector spectral measures of A; their variance splits into a B term and a nonnegative Y term. This is not equivalent to a pointwise inequality $p(A) \geq p(B)$.

The two-eigenvalue statement `mat:scalar-pair` bounds endpoint residuals, including divided differences of k, b, f_*, by

$$Q(G, g - G/2) + r_{ij}\delta_{ij}^2 Z^2, \qquad Q(u,v) = \frac{43}{100}u^2 + \frac{26}{100}uv + \frac{68}{25}v^2.$$

The coefficients of the separation term must agree with (8.5). Summing it cancels $\mathcal{C}$. The remaining chaos computation is independently treated in Section 13. A finite determinant certificate for K does not establish either scalar statement.

9 Scalar certificates: polynomial equivalence is not enough

M203's `certificate.tex` splits the real line into $a \leq -6$, $[-6, 10]$, and $a \geq 10$. Both unbounded tails are treated analytically; the bounded interval uses rational polynomials. The central construction is sufficiently explicit for an independent replay. The independent reconstruction passes the finite checks specified in Section 16; that replay was rerun locally. The distinction between finite comparisons and the analytic arguments around them is retained.

9.1 The exact finite data that must be checked

On each panel $a = w + rx$, $-1 \leq x \leq 1$, the source builds degree-14 polynomials $P, \widehat{k}$. The centers and radii are

w	-5.25	-3.75	-2.5	-1.5	$-.5$	$.25$	1.25	3	5	7	9
r	$.75$	$.75$	$.5$	$.5$	$.5$	$.25$	$.75$	1	1	1	1

and the decimal seeds in `cert:centers` are exact rationals, not exact Gaussian function values. With $D = P - a$, the recurrences are

$$P_{i+1} = \frac{r}{i+1}(PD)_i, \qquad \widehat{k}_{i+1} = \frac{r}{i+1}(1 - \widehat{k}D)_i. \tag{9.1}$$

Their validity as a certificate requires checking the differential residuals, all 20 joining discrepancies, and both analytic anchors. The source states global errors

$$|q - P| \leq 10^{-8}, \qquad |k - \widehat{k}| \leq 8 \cdot 10^{-7} \quad \text{on } [-6, 10]. \tag{9.2}$$

The stable directions are backward for q, from the right anchor, and forward for k, from the left anchor. Recurrences alone do not prove (9.2); rational seeds by themselves have no certified relationship with q or k.

Next form the *full* auxiliary polynomials

$$B_p = 1 - \widehat{k}D, \qquad J_p = \widehat{k} - P + aB_p, \qquad Y_p = 2B_p - PD + aJ_p.$$

They approximate b, l, l', respectively. In the source error calculus, if $[p]$ denotes the sum of absolute power coefficients on $[-1, 1]$, and the errors for q, k are $\varepsilon_q, \varepsilon_k$, one can use

$$\begin{aligned} E_b &= [\widehat{k}]\varepsilon_q + \varepsilon_k(\varepsilon_q + [D]), \\ E_l &= \varepsilon_q + \varepsilon_k + [a]E_b, \\ E_{l'} &= 2E_b + ([P] + [D] + \varepsilon_q)\varepsilon_q + [a]E_l. \end{aligned}$$

Only after these full products are formed does the source truncate to quartics, charging the absolute coefficient tail to the error. For enclosures $u = p + e_u$, $v = q + e_v$, $|e_u| \le E_u$, $|e_v| \le E_v$, a valid product error is

$$[p]E_v + [q]E_u + E_uE_v, \tag{9.3}$$

plus the charged polynomial truncation tail. Dropping high-degree terms before forming the auxiliary expressions would certify a different polynomial.

For a quartic $A(x) = \sum_{j=0}^{4} A_j x^j$ with uniform error E, the source's `cert:Bernstein` is

$$\min_{\sigma\in\{-1,1\},\, 0\le i\le 4} \sum_{j=0}^{i} \frac{\binom{i}{j}}{\binom{4}{j}} \sigma^j A_j - E. \tag{9.4}$$

It is a rigorous lower bound on $[-1, 1]$, by applying (6.3) on each half interval. This coefficient equivalence is exactly grounded by the local volume-16 theorem. The certificate's remaining substance is whether the correct A_j and sufficiently large E have been supplied.

9.2 The stated central margins and tails

The source reports the following minima, in units of 10^{-3}, for four groups of inequalities. This table transcribes the claimed bounds in `cert:group-bounds`; it is not a table of the reconstructed minima. All 44 comparisons were separately replayed successfully, as described in Section 16, without claiming formal verification of the full analytic argument.

w	$\mathcal{G}_1$	$\mathcal{G}_2$	$\mathcal{G}_3$	$\mathcal{G}_4$
-5.25	7	1	1	1
-3.75	1	1	2	2
-2.5	2	2	2	5
-1.5	2	2	2	8
-.5	2	3	5	14
.25	2	2	8	11
1.25	2	.8	8	20
3	2	.8	8	7
5	1	.8	10	15
7	3	.6	11	11
9	2	.35	12	9

The groups cover bounds on b, l, p, h_0, derivative/log-slope bounds, scalar surplus controls, and the leading minor and determinant of a two-by-two quadratic-form comparison. Clearing a denominator requires its separately established positive sign. Checking just the last determinant is not a positive-semidefinite matrix certificate without its leading-minor condition.

For the left tail, the source expands integrals of the form $\int_0^\infty v^j e^{-rv-v^2/2}\,\mathrm{d}v$, with $r = -a$. For the right tail, it uses $\int_0^\infty e^{-v-uv^2/2}\,\mathrm{d}v$, $u = a^{-2}$, and logarithmic corrections. Bounds are propagated

through derivatives up to the required order. A central Bernstein proof cannot replace these tail arguments. Likewise finitely many evaluations of a Gaussian CDF cannot prove the real-line scalar inequalities.

The appropriate independent certificate package would include the exact seed fractions, each residual polynomial and its coefficient norm, each join difference, anchor inequalities with explicit analytic remainders, all auxiliary truncation budgets, every denominator positivity check, and the tail estimates. The missing source verifier is an implementation gap in the inspected package, not by itself a disproof of the mathematical argument.

10 Stable residual propagation with rational damping

This section proves a refinement of the *error propagation step*. It does not certify that M203's input residual numbers are correct.

Theorem 10.1 (Backward and forward scalar error transport)**.** *Let $q' = q(q-a)$ and $k' = 1-k(q-a)$ on an interval. Let differentiable approximants $P, \widehat{k}$ have residuals*

$$R_q = P' - P(P-a), \qquad R_k = \widehat{k}' - (1 - \widehat{k}(P-a)).$$

Write $e = q - P$, $z = k - \widehat{k}$, $d = q - a$. Then

$$e' = (q + P - a)e - R_q, \qquad z' = -dz - \widehat{k}e - R_k. \tag{10.1}$$

If $q + P - a \geq \lambda > 0$ on $[a_0, a_1]$, with $h = a_1 - a_0$, and $|R_q| \leq \rho_q$, then

$$|e(a_0)| \leq e^{-\lambda h}|e(a_1)| + \rho_q \frac{1 - e^{-\lambda h}}{\lambda}. \tag{10.2}$$

If $d \geq \nu > 0$, $|\widehat{k}| \leq M$, $|e| \leq E_q$, and $|R_k| \leq \rho_k$ throughout the interval, then

$$|z(a_1)| \leq e^{-\nu h}|z(a_0)| + (\rho_k + ME_q)\frac{1 - e^{-\nu h}}{\nu}. \tag{10.3}$$

At a switch to a neighboring approximating polynomial, its known joining discrepancy must be added to the propagated endpoint bound.

Proof. Subtract the approximate equations from the exact ones: $q(q-a) - P(P-a) = (q + P - a)(q - P)$, and

$$-k(q-a) + \widehat{k}(P-a) = -(q-a)(k - \widehat{k}) - \widehat{k}(q-P).$$

Variation of constants for the first equation, integrated backward, gives

$$e(a_0) = e^{-\int_{a_0}^{a_1}(q+P-v)\,\mathrm{d}v} e(a_1) + \int_{a_0}^{a_1} e^{-\int_{a_0}^{t}(q+P-v)\,\mathrm{d}v} R_q(t)\,\mathrm{d}t.$$

Taking absolute values and using the lower coefficient bound proves (10.2). The forward integrating factor for the second equation proves (10.3). Changing approximants changes the error by their difference, proving the join rule. □

For the source panels, $q > 0$ and a certified $P - a > \lambda$ suffice for the backward coefficient condition; no prior tight error bound on $q - P$ is needed for that condition. A lower bound for d can be obtained from $P - a - E_q$, if positive. Residuals computed in $x = (a - w)/r$ must be divided by r to become a-derivative residuals; the panel length is $2r$. Confusing these two coordinates changes the error budget.

Lemma 10.2 (Rational upper damping factors)**.** *For $x \geq 0$,*

$$e^{-x} \leq \frac{1}{1+x}, \qquad \frac{1-e^{-x}}{x} \leq \frac{1}{1+x/2}, \tag{10.4}$$

where the second left side is one at $x = 0$.

Proof. The first inequality follows from $e^x \geq 1+x$. For $x \geq 2$, the second follows from $(1-e^{-x})/x \leq 1/x \leq 1/(1+x/2)$. For $0 \leq x < 2$, differentiate $\log((1-x/2)/(1+x/2)) + x$. Its derivative is $1 - 1/(1-x^2/4) \leq 0$, and it is zero at $x = 0$. Thus $e^{-x} \geq (1-x/2)/(1+x/2)$, which rearranges to the result. □

For example (10.2) admits the completely rational update

$$E_{\text{left}} \leq \frac{E_{\text{right}}}{1+\lambda h} + \frac{\rho_q h}{1+\lambda h/2}. \tag{10.5}$$

The analogous forward update uses ν and $\rho_k + ME_q$. These are no larger than the undamped bound $E + \rho h$ and can be strictly smaller. The interval-wide E_q used in a k update must bound the error throughout the panel, not just one endpoint. This refined propagation can reduce certified approximation budgets without altering the polynomial coefficient identities or claiming a new Gaussian inequality.

11 A complete raw-moment entropy certificate

A moment determinant alone cannot decide the entropy inequality for a general density. There is an exact additional scalar integral.

Theorem 11.1 (Entropy from mass, energy, and augmented moments)**.** *Let $V : \mathbb{R}^m \to (-\infty, \infty]$ be measurable, set $w = e^{-V}$, and suppose*

$$0 < Z = \int w < \infty, \qquad E_V = \int Vw \text{ is absolutely convergent,}$$
$$H_V = \int (1,x)(1,x)^T w(x)\,\mathrm{d}x \succ 0$$

with finite entries. Use $Vw = 0$ where $V = +\infty$. For $f = w/Z$,

$$\Delta_m(f) = \frac{E_V}{Z} - m + \frac{m+3}{2}\log Z - \frac{1}{2}\log\det H_V. \tag{11.1}$$

Suppose certified enclosures give $0 < z_- \leq Z \leq z_+$, $e_- \leq E_V \leq e_+$, and $0 < d_- \leq \det H_V \leq d_+$. Let a_- and a_+ be the minimum and maximum of the four numbers e_σ/z_τ, $\sigma, \tau \in \{-, +\}$. Then

$$\Delta_m(f) \geq a_- - m + \frac{m+3}{2}\log z_- - \frac{1}{2}\log d_+, \tag{11.2}$$

$$\Delta_m(f) \leq a_+ - m + \frac{m+3}{2}\log z_+ - \frac{1}{2}\log d_-. \tag{11.3}$$

Proof. The identity $-\log f = V + \log Z$ gives $h(f) = E_V/Z + \log Z$. The Schur-complement calculation of Proposition 4.1, with mass Z in place of volume, gives $\det \mathrm{Cov}(f) = \det H_V / Z^{m+1}$. Substitution proves (11.1). A ratio with positive denominator has its extremum over this rectangle at a corner. The logarithms are monotone, giving the interval inequalities. □

This theorem does not need convexity for the algebra; convexity of V is an extra hypothesis needed to regard a passing test as an instance of the log-concave target. The corner rule handles negative energy integrals correctly. It would be unsafe to divide e_- by z_+ automatically when e_- is negative.

Potential shifts provide a useful check. Replacing V by $V + c$ changes

$$Z \mapsto e^{-c} Z, \quad H_V \mapsto e^{-c} H_V, \quad E_V \mapsto e^{-c}(E_V + cZ).$$

The added c, the mass logarithm and the determinant logarithm cancel exactly in (11.1). Thus the formula depends on the normalized density, as it must.

For rational logarithm bounds, $x > 0$, $v = (x-1)/(x+1)$, and $N \geq 0$,

$$\log x = 2 \sum_{j=0}^{N} \frac{v^{2j+1}}{2j+1} + R_N, \qquad |R_N| \leq \frac{2|v|^{2N+3}}{(2N+3)(1-v^2)}.$$

This follows by integrating the geometric series for $1/(1-v^2)$. All terms and the remainder bound are rational for rational x. Rescaling toward one accelerates convergence but is not necessary for validity.

For an unbounded density every mass, moment and energy enclosure must include tails. A finite box quadrature with no tail control does not meet the theorem's hypotheses. A positive lower bound from (11.2) certifies that specified density. A rigorously negative upper bound from (11.3), together with verified log-concavity and dimension assumptions, would be a genuine counterexample to the target, not merely an unfavorable coordinate residual. No such counterexample is produced here.

12 Local rigidity and quantitative curl jets

The source's equality mechanism is stronger than matching finitely many moments. It uses compatibility of an actual nonlinear Jacobian. This mechanism admits a precise quantitative local formulation.

Theorem 12.1 (Curl-jet control of coefficient commutators)**.** *Let $X_1, \dots, X_m \in \mathrm{Sym}_m(\mathbb{R})$, $A(x) = \sum_r x_r X_r$. Let g be analytic near zero with $g'(0)g''(0) \neq 0$. Define $J(x) = g(A(x))$ near zero and*

$$C_{rs}(x) = (\partial_r J(x)) e_s - (\partial_s J(x)) e_r, \qquad E_{rs} = X_r e_s - X_s e_r.$$

Then

$$C_{rs}(0) = g'(0) E_{rs}, \tag{12.1}$$

$$\frac{2}{g''(0)} \partial_j C_{rs}(0) = [X_r, X_s] e_j + X_r E_{js} - X_s E_{jr} + X_j E_{rs}. \tag{12.2}$$

If $\|X_r\|_{\mathrm{op}} \leq L$, $\|C_{rs}(0)\| \leq \varepsilon_0$, and $\|\partial_j C_{rs}(0)\| \leq \varepsilon_1$ for all indices, then

$$\|[X_r, X_s]\|_F \leq \sqrt{m} \left(\frac{2\varepsilon_1}{|g''(0)|} + \frac{3L\varepsilon_0}{|g'(0)|} \right). \tag{12.3}$$

If $\sum_r X_r^2 = I$, one may take $L = 1$.

Proof. The convergent matrix series gives

$$J = g(0)I + g'(0)A + \tfrac{1}{2}g''(0)A^2 + O(|x|^3).$$

The constant curl term is (12.1). Its degree-one coefficient is

$$\frac{2}{g''(0)}\partial_j C_{rs}(0) = (X_r X_j + X_j X_r)e_s - (X_s X_j + X_j X_s)e_r.$$

Use $X_j e_s = X_s e_j + E_{js}$, $X_j e_r = X_r e_j + E_{jr}$, and $X_r e_s - X_s e_r = E_{rs}$ to obtain (12.2). Bound the three error terms in norm and then sum the squared bounds over the m columns of the commutator. Finally $X_r^2 \preceq \sum_s X_s^2 = I$ for symmetric X_r, which implies $\|X_r\|_{\mathrm{op}} \le 1$. □

This is a finite-jet result. An L^2 curl error does not by itself control the pointwise jets in its hypotheses. Additional regularity and a proved evaluation bound would be required to use (12.3) with an integral numerical residual.

For $g = q$ from (8.1),

$$q'(0) = 2/\pi, \qquad q''(0) = \sqrt{2/\pi}\,(4/\pi - 1) > 0.$$

If $q(A)$ is an actual weak Jacobian, its distributional curl vanishes. Near zero these curl functions are analytic, so the zeroth and first jets vanish pointwise. The theorem gives $[X_r, X_s] = 0$. Simultaneous orthogonal diagonalization, with the same rotation of input and output coordinates, then makes the X_r diagonal. The condition $E_{rs} = 0$ forces only the r-th diagonal entry of X_r to survive. If $\sum_r X_r^2 = I$, each surviving entry is a sign σ_r. Integrating the Jacobian gives

$$U^T F(Uy) = (\sigma_i t(\sigma_i y_i))_{i=1}^m + w.$$

Since $\Phi(\sigma_i G_i)$ are independent uniform random variables, $t(\sigma_i G_i)$ are independent mean-one exponentials. This proves the local-to-global algebraic implication in `lem:linear-jacobian`, given the globally compatible linear field.

The nonzero second derivative is substantive. With $g(a) = a$, choose

$$X_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $A(x) = x_1 X_1 + x_2 X_2$ is the Hessian of $x_1^3/3 + x_1 x_2^2/2$, so it is a compatible Jacobian, but

$$[X_1, X_2] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0.$$

This example is not a positive global transport Jacobian and is not a counterexample to M203. It shows why linear compatibility alone does not replace the nonlinear two-jet argument.

12.1 Why vanishing entropy slack is a separate step

The source first needs its quantitative smooth remainder to make $Y_j \to 0$ and $B_j \to I$ along an equality-approximating sequence. After finite-dimensional compactness of the $X_{r,j}$, one has $A_j \to A = \sum_r x_r X_r$ in Gaussian L^2. A direct two-eigenbasis calculation proves

$$\|q(M) - q(N)\|_F \le \|M - N\|_F \quad (M, N \text{ real symmetric}),$$

because q is scalar 1-Lipschitz. Thus $J_j \to q(A)$ in Gaussian L^2, and Gaussian Poincare for centered transport maps gives strong H^1 convergence. The weak limiting Jacobian is compatible and the preceding rigidity argument applies.

This is a Frobenius-norm statement, not the generally false rule that every scalar Lipschitz function is operator-Lipschitz in operator norm. It is also not a claim of uniform Hessian bounds for the entire approximating sequence. Only *after* identifying the limit covariance as I does the source use $P_jC_jP_j \to I$, together with $C_j \to C \succ 0$, to control the normalizing matrices and their inverses. Reversing that order would introduce an unproved compactness assumption.

13 Exact chaos slack and stronger conditional constants

The final spectral step can be proved independently of the transport problem. All norms in this section are Gaussian L^2, with Frobenius summation over matrix entries.

Theorem 13.1 (Exact spectral remainder)**.** *Let $Y \in H^1(\gamma_m)$ be a matrix-valued field with no degree-zero or degree-one Hermite chaos. Write $Y = \sum_{d\ge 2} Y_d$, $G = \partial Y$, $g = \partial N^{-1}Y$, and $V_0 = \|G\|^2 - \|Y\|^2$. For the quadratic form Q above,*

$$V_0 - \mathbb{E}\sum_{i,j,r} Q\big((G_r)_{ij}, (g_r - G_r/2)_{ij}\big) - \frac{7}{50}\|Y\|^2 = \sum_{d\ge 2} \frac{(d-2)(d+68)}{50d}\|Y_d\|^2. \tag{13.1}$$

The series is finite and nonnegative.

Proof. For degree d, $g = G/d$ and $\|G\|^2 = d\|Y_d\|^2$. Orthogonality is preserved under differentiation between distinct total degrees. Direct rational algebra gives the polynomial identity

$$1 - u - Q(1, u - \tfrac{1}{2}) - \frac{7}{50}u = \frac{1}{50}(1-2u)(1+68u). \tag{13.2}$$

Set $u = 1/d$, multiply by $d\|Y_d\|^2$, and sum. The coefficient is $O(d)$, so the H^1 assumption ensures convergence. □

Denote the right side of (13.1) by R_{ch}. It vanishes at degree two. For $d \ge 3$ its coefficient is increasing and at least $71/150$. Thus it gives meaningful additional control of chaoses of degree at least three, but no improvement for a pure degree-two perturbation.

If M203's upstream surplus, ordered expansion, absorption and two-eigenvalue inequalities hold, `mat:pre-pair` and its endpoint summation yield

$$\mathcal{D} \le -V_0 + \frac{69}{500}\|Y\|^2 - \frac{\Omega_B}{1000} + \mathbb{E}\sum Q(G, g - G/2).$$

The independently proved spectral identity then yields the conditional improvement

$$2\Delta_m(f) \ge \frac{1}{500}\|Y\|^2 + \frac{1}{1000}\Omega_B + R_{\mathrm{ch}}.$$

The rest of this section sharpens the first two constants as well. It incorporates the independent source audit, Section 10, with its rational derivation checked against the source's `scalar.tex`.

13.1 Removing three rounding losses

Set, exactly,

$$y_0 = \frac{40}{37}\left[\left(\frac{7}{25}\right)^2 + \frac{(3/40)(67/100)^2}{4(36/100)}\right], \tag{13.3}$$

$$r_0 = \frac{(64/100)(128/100)^2(13/100)}{16(46/100 - 13/100)}. \tag{13.4}$$

The source's `sc:y-cost` bounds the first Y^2 cost by y_0, then rounds it up to 0.111. Keep y_0 instead.

For an eigenvalue λ of B, let

$$u = \min\{.13, 1.28/\sqrt{\lambda}\}, \qquad d_u = \frac{u^3}{16(.46 - u)}.$$

The positive quadratic-moment loss in `sc:rho-loss` is at most r_0: for $\lambda \geq 1$, use $H_* \leq \lambda$, $\rho \leq .64$, $\lambda u^2 \leq 1.28^2$, and $u \leq .13$; for $\lambda \leq 1$, use $H_* \leq \alpha s$, which gives the smaller bound $\alpha s(.64)(.13)^3/[16(.46 - .13)]$. There is no need to round the common bound to 0.027.

The same source minorant gives, for the actual vector spectral measure ν with second moment $\lambda + r$,

$$\int p \, \mathrm{d}\nu \geq \frac{569}{825}u - d_u r, \quad \frac{569}{825} = 1 - \frac{1.28^2}{16(.46 - .13)}. \tag{13.5}$$

The two already established bounds in `sc:log-bound` give

$$\frac{\alpha F_0(S)}{\rho} \leq \min\{.089, .88/\sqrt{\lambda}\} \leq \frac{11}{16}u,$$

because $.089/.13 < 11/16$ and $.88/1.28 = 11/16$. At $\lambda = 0$ the constant branch is used, and $\lambda = 1$ is harmless by continuity.

Consequently the absorption estimate strengthens to

$$[\text{left side of source } \texttt{mat:absorption-inequality}] \leq (y_0 + r_0)\|Y\|^2 - \zeta\Omega_B, \qquad \zeta = \frac{569}{825} - \frac{11}{16} = \frac{29}{13200}.$$

Combining with the same pair estimate and Theorem 13.1 leaves

$$\eta = \frac{7}{50} - y_0 - r_0 = \frac{1689103}{407000000} > \frac{1}{250}, \qquad \zeta = \frac{29}{13200} > \frac{1}{500}. \tag{13.6}$$

These arithmetic improvements do not change any central scalar panel, two-eigenvalue constant, or equality class. They improve a quantitative smooth remainder, not the leading constant m in the entropy target.

13.2 Two additional positive terms

There are two further exact terms that the source discards. At `tr:initial-multiplier`, retain

$$R_V = \mathbb{E}\,\mathrm{tr}\left(t(A)^2 D^2 V(F)\right) \geq 0.$$

Indeed the discarded term is $\mathbb{E}\,\mathrm{tr}(k(A)^2 J D^2 V(F) J)$; trace cyclicity and $Jk(A) = t(A)$ give precisely R_V, with no commutation assumption on $D^2 V(F)$. Keeping it adds R_V to the right side of (8.5).

For the log-determinant supporting step, define

$$R_S = \operatorname{tr}(SC) - \log\det(S^{1/2}CS^{1/2}) - m \geq 0.$$

The nonnegative scalar function $x - 1 - \log x$, applied to the positive eigenvalues of $S^{1/2}CS^{1/2}$, proves this sign. The exact supporting identity is

$$\log\det C + m = \operatorname{tr}(SC) - \log\det S - R_S.$$

One must use the positive congruence, not pretend SC is symmetric.

Corollary 13.2 (Strengthened smooth remainder, conditional)**.** *Assume the normalized smooth transport setting of Section 8 and the source's analytic surplus and scalar absorption/pair inequalities, with their stated domains and regularity justifications. Then its proof, with the displayed terms retained, gives*

$$\boxed{2\Delta_m(f) \geq R_V + R_S + R_{\mathrm{ch}} + \frac{1689103}{407000000}\|Y\|^2 + \frac{29}{13200}\Omega_B.} \tag{13.7}$$

Proof. Keep R_V in the transport surplus and R_S in the supporting identity. They enter the signed deficit bound with negative signs. Replace the source's rounded absorption cost by (13.3)–(13.5); its ordered matrix estimates and pair cancellation are otherwise unchanged. Apply the exact identity (13.1) and use (13.6). □

This is a rigorous implication with identified premises, not an independent proof that all those premises hold for every transport. All terms are finite in the fixed smooth two-sided-Hessian setting. There $R_V \geq c_-\mathbb{E}\operatorname{tr}(t(A)^2) > 0$, using the Hessian lower bound in the *normalized* coordinates. Thus equality cannot occur in that strictly uniformly convex class, consistent with the one-sided exponential equality laws lying at a nonsmooth boundary. No termwise passage of (13.7) to arbitrary nonsmooth densities or convex bodies has been proved in this report.

14 Approximation and geometric limits

The source does not stop at a strictly convex smooth class. For a proper lower semicontinuous convex potential V with $0 < \int e^{-V} < \infty$, `geo:coercivity` derives a global lower bound $V(x) \geq a|x| - b$, $a > 0$. The reason is geometric: an appropriate finite-volume full-dimensional convex sublevel set is bounded, and convexity converts its radial growth to a linear lower bound. This also supplies exponential control of moments and, after a modest weakening of the exponent, of $|V|e^{-V}$.

The source uses the Moreau envelope

$$Q_j(x) = \inf_y\{V(y) + j|x-y|^2/2\}, \qquad V_j = Q_j * \eta_{1/j} + |x|^2/j,$$

where the mollifier is smooth, nonnegative, even, and compactly supported. It obtains

$$\frac{2}{j}I \preceq D^2V_j \preceq (j + 2/j)I, \qquad V_j(x) \geq a|x| - B$$

with a common tail bound. Pointwise convergence where appropriate, together with these tails, gives convergence of normalization, L^1 density, entropy, mean, and covariance. Weak convergence alone would not justify entropy convergence.

These approximation results transfer a proved smooth inequality to the full log-concave class. They do not supply the missing sign if the smooth inequality has not been established. For equality, Section 12's compactness mechanism is additional. In particular, the lower Hessian constants $2/j$ tend to zero, so an error estimate requiring a uniform strictly positive lower Hessian bound cannot simply be carried to the limit.

On the body side, TR-item-344 supplies a different continuity route. For $K \subset L \subset c + t(K - c)$, $t \geq 1$, its `thm:continuity` gives

$$t^{-n(n+3)} \leq \frac{F(L)}{F(K)} \leq t^{n(n+3)}.$$

For inner/outer Hausdorff approximation at scale $\delta < r$, where r is an inradius, a homothety parameter can be $t = (1 + \delta/r)/(1 - \delta/r)$. Hence certified rational polytopes can approximate a specified body with a complete geometric error budget. Countable density of rational polytopes is not a finite covering certificate for all convex bodies, and it does not establish a uniform positivity margin near a simplex equality case.

15 Facet residuals: a relevant alternative with a strict boundary

The variational part of TR-item-344 is a second core route to the body problem, not needed for M203's entropy proof. In isotropic coordinates $\mu = 0, \Sigma = I$, a two-sided admissible facet variation with velocity $c + w \cdot x$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{K_t} p(x)\,\mathrm{d}x \Big|_{t=0} = \int_F (c + w \cdot x) p(x)\,\mathrm{d}x.$$

After differentiating $\log F$, stationarity yields the facet residual

$$r_F = \mathbb{E}_F\big[(|X|^2 - (n+2))X\big] = 0$$

for the admissible variations used in `lem:variation`. The expectation is normalized surface measure on the actual facet. This is not valid for an arbitrary vector called a facet moment.

For the adjacent-facet setup of `thm:reflection`, choose coordinates so the linear map fixing the ridge and the origin has $A = [e_1, \ldots, e_{n-1}, v]$, $v_n < 0$. Let

$$M = \mathbb{E}_F[X_n X X^T] \succ 0, \qquad w = \big(v_{\tan}, (|v|^2 - 1)/2\big).$$

The source's exact identity is

$$2Mw = A^{-1} r_G - r_F. \tag{15.1}$$

With a certified $\lambda_{\min}(M) \geq \mu > 0$, $\|A^{-1}\| \leq B_*$, and $\|r_F\| \leq \varepsilon_F$, $\|r_G\| \leq \varepsilon_G$, this gives

$$\|w\| \leq d_* = \frac{B_* \varepsilon_G + \varepsilon_F}{2\mu}, \qquad \|A - R_0\| \leq \sqrt{d_*^2 + (2d_* + d_*^2)^2},$$

where R_0 is the corresponding exact reflection in those coordinates. A lower eigenvalue certificate is indispensable; a tiny residual with nearly singular M can have a large geometric effect.

This is a concrete use of weighted moment/Gram certificates. It does not give global simplex rigidity. The isotropically normalized cube has these facet stationarity residuals zero and exact facet reflections, yet is not a simplex for $n \geq 2$ and has $F = 12^{-n} < S_n$. Thus even exact local symmetry data do not identify the global maximizer. A future bridge must use additional extremality or global geometric information, not merely tighten residual arithmetic.

16 Independent audit integration and verification record

The completed independent source audit was read in full at

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M203-source-audit.txt`.

Its Sections 4–9 review the analytic and algebraic chain; Section 10 derives the improved constants; Section 11 identifies the additional positive remainders. Its own depth declaration is important: the external transport existence and contraction theorems were not independently reproved or checked against all original publications, and the analytic tail, Sobolev, degree and compactness arguments were mathematically reviewed, not machine-formalized. It reports no demonstrated fatal error within that bounded audit.

Both audit helper sources were read, and both were rerun locally with the bundled Python interpreter:

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M203_scalar.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M203_matrix.py
```

They were left unchanged. These are independent reconstructions from the printed source, not the source author's missing verifier.

The scalar helper uses exact rational polynomial arithmetic. The local rerun returned `CHECKS 201; FAILURES 0`. Its coverage includes all 44 central group comparisons, 22 residual bounds, 20 joins, 11 lower bounds for $P - a$, 11 coefficient-norm bounds for $\widehat{k}$, endpoint and Mills-anchor comparisons, accumulated error bounds, logarithmic separation bounds, 36 substitution corner comparisons, selected tail majorants, and the improved constants. The assertions are not 201 independent analytic theorems: several are finite comparisons within the same enclosing argument.

For example, the minimum recomputed $\mathcal{G}_2$ bound at center 9 is approximately 0.000508258581455, above the claimed 0.00035. At center -3.75, the recomputed $\mathcal{G}_1$ minimum is approximately 0.00135441835944, above the claimed 0.001. The decisions are rational comparisons; these decimals are only readable summaries. The source table in Section 9 remains a table of the *claimed* lower bounds, now supported by this separately reconstructed finite replay.

The matrix helper returned exact ordered-expansion discrepancy zero, exact commutator-square discrepancy zero, and a passing commutator inequality in its noncommuting polynomial stress case. It uses

$$X_1 = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}, \quad B = \operatorname{diag}(6, 9),$$

nonzero quadratic and cubic Gaussian chaoses, and an artificial polynomial $k(a) = 1+2a+a^2/2+a^3/6$. This checks transpose-sensitive algebra after a stated scalar substitution; it is *not* a Gaussian-hazard test, an actual Brenier transport, or a test of log-concavity. The general commutator lemma is proved independently in Proposition 8.1; a single stress example is not being used to prove a universally quantified matrix inequality.

The companion `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M203-Isotropic-Constants/verify_refinements.py` checks this report's exact finite algebra: simplex determinants, the separated relative-bound example, the chaos polynomial, the constants in (13.6), generic two-dimensional curl-jet identities, Bernstein conversion, and covariance/entropy normalization identities. These checks complement the proofs; they do not run or certify the entire transport argument. The local execution returned `CHECKS 133; FAILURES 0`. Its formal noncommutative-word checks cover the curl identity in dimensions two and three; the dimension-independent proof remains Theorem 12.1.

A subsequent independent review of the mass-coupled, covariance-Gram, rational-damping, raw-entropy, curl-jet and compensated-commutator results identified the need to state the integer cutoff in Theorem 7.1 explicitly. That correction and its noninteger counterexample are now included. The review found no further substantive issue in those six results under their stated hypotheses. Its concise checks are appended to the same exact source-audit path above.

16.1 What the audit changes, and what it does not

Before the independent replay, the central scalar margins were unverified source numbers at the initial reading stage. They are now supported by a separate exact implementation and a successful local rerun. That is materially stronger evidence for this finite component. It does not eliminate the distinction between polynomial arithmetic and the analytic premises that connect it to actual transports. The report's conditional labeling of Corollary 13.2 is therefore retained.

The exact source paths and labels above distinguish this report's deductions from source assertions. The confidential corpus was examined locally and was not uploaded for the public literature checks.

17 Advantages, limitations, and concrete next work

17.1 What is gained

The Ram connection is unusually direct in this manuscript: TR-item-344 uses exactly the body functional in (1.1), and the cone identity connects it exactly to the source's functional deficit. Physical Gram coordinates make positivity, affine congruence, and full residual accounting explicit. The polynomial bank and Bernstein conversion give a reproducible finite representation of the same scalar certificate that M203 needs.

The mass-coupled certificate improves a practical robust test with a complete proof and an exact separating example. The covariance Gram-tail theorem identifies exactly what a finite transport coefficient computation would need to enclose. The raw-moment entropy formula prevents covariance-only reasoning from being mistaken for an entropy calculation. The curl-jet theorem adds a quantitative local compatibility check. Finally, (13.7) improves the source's smooth remainder by retaining already available arithmetic and analytic slack.

17.2 What remains costly or unavailable

There is no general theorem that a truncated positive moment matrix is a uniform convex-body moment matrix. There is no finite-dimensional surrogate for the nonlinear degree normalization supplied by merely changing polynomial coordinates. Finite Gaussian coefficients require integration and tails; the spectral degree count itself grows rapidly with dimension. Scalar interval certificates require exact error budgets and complete unbounded-tail arguments.

Near equality, strictly positive Bernstein margins can disappear. The smooth curvature remainder need not survive the nonsmooth limit with a useful quantitative lower bound. Small higher-chaos energy and small Ω_B do not yet provide an explicit distance to an exponential product or to a simplex. Finite-jet curl bounds need pointwise derivative control, which is stronger than the Gaussian L^2 convergence used for qualitative equality. The public Mahler implication was not independently audited.

17.3 Actionable next steps

1. Preserve the reconstructed scalar certificate as exact rational data with an explicit input manifest and independent implementation comparison. Check the full tail-to-central interface, not only the 44 central margins. The completed replay already supplies a sound starting point.
2. Audit the imported full-space transport regularity and inverse contraction hypotheses against primary originals. Then isolate the normalization map continuity and compact-zero-set proof in a self-contained lemma. This is a precise remaining source-chain task, not a request to search for an unrelated transform theorem.
3. Formalize the ordered matrix expansion and weighted completion in arbitrary dimension. Preserve the H^1–H^{-1} domain, right versus left scalar weights, transposes, and nullspace pseudoinverses. Proposition 8.1 can serve as an independently proved component.
4. Implement the mass-coupled test on verified rational polytope families, including exact triangulation and denominator signs. Report a family certificate, with its parameter box, rather than an all-body conclusion.
5. For a specified smooth log-concave density, enclose the exact mass, energy and augmented moments from Theorem 11.1. Independently compare its deficit with a certified Gaussian transport Gram truncation. This supplies cross-checks with genuinely different error sources.
6. Develop a quantitative passage from the normalized transport remainders to distance from an exponential product. Only after proving the required compactness and stability estimates should the cone identity be used to seek a body-distance theorem.

No further global conjecture claim follows simply by composing the maps in the corpus. The useful outcome is a better exact certificate framework, several proved refinements, and a substantially more specific audit of the source's actual core operator.

References

[1] OpenAI, *A sharp entropy bound and the simplex inequality for isotropic constants*, manuscript dated October 5, 2026. Public manuscript at the pinned commit: `https://github.com/openai/math/blob/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-sharp-entropy-bound-and-the-simplex-inequality-for-isotropic-constants-October-5-2026/isotropic-simplex.pdf`. The exact local-copy path and source labels are in Section 2. Source claims are distinguished from the results proved in this report.

[2] *TR-item-344-astra*, local technical report, September 26, 2026. Confidential local source: `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-344-astra.tex`. Exact theorem anchors and scope are listed in Section 2.1.

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[4] M. Fradelizi and F. Marín Sola, *Entropy, slicing problem and functional Mahler's conjecture*, public preprint, arXiv:2406.07406v1 (2024). `https://arxiv.org/html/2406.07406v1`. Conjecture 3(iii), Proposition 5.2 and Lemma 5.2 consulted. The local M203 bibliography lists a 2026 journal version; that publication metadata was not independently confirmed here.

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